

Reducing the Height of Independent Spanning Trees in Chordal Rings

Jinn-Shyong Yang, Jou-Ming Chang, Shyue-Ming Tang, and Yue-Li Wang

Abstract—This paper is concerned with a particular family of regular 4-connected graphs, called chordal rings. Chordal rings are a variation of ring networks. By adding two extra links (or chords) at each vertex in a ring network, the reliability and fault-tolerance of the network are enhanced. Two spanning trees on a graph are said to be independent if they are rooted at the same vertex, say, r , and for each vertex $v \neq r$, the two paths from r to v , one path in each tree, are internally disjoint. A set of spanning trees on a given graph is said to be independent if they are pairwise independent. Iwasaki et al. [9] proposed a linear time algorithm for finding four independent spanning trees on a chordal ring. In this paper, we give a new linear time algorithm to generate four independent spanning trees with a reduced height in each tree. Moreover, a complete analysis of our improvements on the heights of independent spanning trees is also provided.

Index Terms—Chordal rings, interconnection networks, fault-tolerant broadcasting, independent spanning trees, internally disjoint path.

1 INTRODUCTION

CHORDAL rings (also called *distributed loop networks*) are a variation of ring networks. By adding two extra links (or chords) at each vertex in a ring network, the reliability and fault-tolerance of the network are enhanced [1], [2], [6], [7]. A number of problems on chordal rings have been studied in the past two decades, including the diameter problem [1], the shortest paths problem [3], the routing, and the fault-tolerant routing problems [10], [11], [12], [15]. A chordal ring $CR(N, d)$ is a graph with vertex set $V = \{0, 1, \dots, N-1\}$ and edge set $E = \{(u, v) | [v - u]_N = 1 \text{ or } d\}$, where $[x]_N$ denotes x modulo N . To ensure that every vertex has four adjacent vertices, we assume $d < N/2$. An example of a chordal ring with $N = 14$ and $d = 4$ is shown in Fig. 1.

Two paths P and Q are *internally disjoint*, denoted by $P \parallel Q$, if they have no common vertex except the two end vertices. A *spanning tree* of a graph G is a subgraph of G that contains all vertices in G and forms a tree. Two spanning trees of G are said to be *independent* if they are rooted at the same vertex, say, r , and, for each vertex $v \neq r$, the two paths from r to v , one path in each tree, are internally disjoint. A set of spanning trees with respect to a given graph is independent if they are pairwise independent. For example, four independent spanning trees of $CR(14, 4)$ are shown in Fig. 2.

A rooted spanning tree in the underlying graph of a network can be viewed as a broadcasting scheme for data communication. Thus, the design of multiple independent spanning trees has applications to the reliable communication protocols. In recent years, the problem of finding independent spanning trees of a given graph has received much attention [4], [5], [8], [14]. Itai and Rodeh [8] first gave a linear time algorithm for finding two independent spanning trees rooted at an arbitrary vertex in a biconnected graph. Cheriyan and Maheshwari [4] showed that, for any 3-connected graph $G = (V, E)$ and for any vertex $r \in V$, three independent spanning trees rooted at r can be constructed in $O(|V|^2)$ time. In [14], Zehavi and Itai conjectured that any k -connected graph has k independent spanning trees rooted at an arbitrary vertex. Recently, Curran et al. [5] presented an $O(|V|^3)$ time algorithm for finding four independent spanning trees rooted at any given vertex in a 4-connected graph. This result has contribution to Zehavi and Itai's conjecture. However, the conjecture is still open for arbitrary k -connected graphs with $k \geq 5$.

Although chordal rings discussed here are 4-connected, the design of efficient algorithms for finding independent spanning trees in chordal rings is still valuable. In [9], Iwasaki et al. gave a linear time algorithm to construct four independent spanning trees in chordal rings. Obviously, reducing the height of a spanning tree rooted at the source node is useful to design an efficient broadcasting scheme. We shall present a new algorithm to generate four independent spanning trees with a reduced height in each tree for chordal rings.

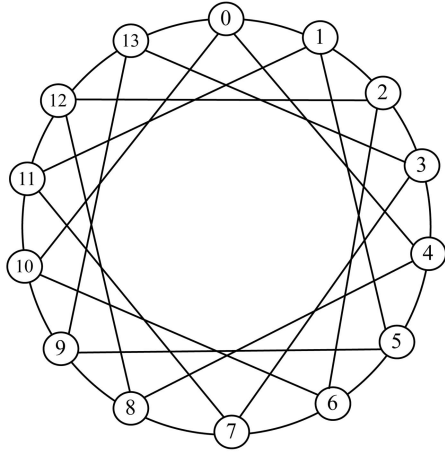
The remaining part of this paper is organized as follows: Section 2 gives a brief description of Iwasaki's algorithm and introduces some prerequisite properties. Section 3 and Section 4 present our algorithms to generate height-reduced independent spanning trees in chordal rings for cases $[N]_d \neq 0$ and $[N]_d = 0$, respectively. To show the correctness of the proposed algorithms, the proof is given by dropping some of the tediously algebraic computations and mentioning only the basic guideline. Section 5 provides a complete analysis of our improvements on the heights of independent

- J.-S. Yang is with the Department of Information Management, National Taiwan University of Science and Technology, Taipei, Taiwan, ROC. E-mail: shyong@mail.ntub.edu.tw.
- J.-M. Chang is with the Department of Information Management, National Taipei College of Business, Taipei, Taiwan, ROC. E-mail: spade@mail.ntub.edu.tw.
- S.-M. Tang is with the Department of Psychology, Fu-Hsing-Kang College, Taipei, Taiwan, ROC. E-mail: joseph.stang@msa.hinet.net.
- Y.-L. Wang is with the Department of Computer Science and Information Engineering, National Chi Nan University, Nantou, Taiwan, ROC. E-mail: ylwang@cs.ntust.edu.tw.

Manuscript received 7 Oct. 2005; revised 27 Apr. 2006; accepted 29 June 2006; published online 8 Jan. 2007.

Recommended for acceptance by Y. Pan.

For information on obtaining reprints of this article, please send e-mail to: tpsds@computer.org, and reference IEEECS Log Number TPDS-0422-1005. Digital Object Identifier no. 10.1109/TPDS.2007.1017.


 Fig. 1. A chordal ring $CR(14, 4)$.

spanning trees. Finally, a concluding remark is given in the last section.

2 IWASAKI'S ALGORITHM AND PREREQUISITE PROPERTIES

Chordal rings are vertex-symmetric [6]. Without loss of generality, any algorithm for constructing independent spanning trees in a chordal ring can simply consider $r = 0$ as the root. Let T_1, T_2, T_3 , and T_4 be four spanning trees of a given chordal ring $CR(N, d)$. For convenience, we write $p(x)$ as the parent of a vertex $x \neq 0$ in a tree. Since the four neighbors of a vertex x in $CR(N, d)$ are $[x + 1]_N$, $[x - 1]_N$, $[x + d]_N$, and $[x - d]_N$, each of the vertices $1, d, N - 1$, and $N - d$ can be assigned as the only child of the root in T_1, T_2, T_3 , and T_4 , respectively. Assume that $2 \leq d < N/2$. The following two procedures are used in [9] for constructing T_1 and T_2 :

Procedure $T_1(N, d)$

begin

1. **for** $x = 1$ **to** $d - 1$ **do**
 $p(x) \leftarrow x - 1$;
2. **for** $x = d$ **to** $N - d$ **do**
if $[x]_d = 0$ **then**
 $p(x) \leftarrow x - 1$;
else
 $p(x) \leftarrow x - d$;
3. **for** $x = N - d + 1$ **to** $N - 1$ **do**
 $p(x) \leftarrow [x + d]_N$;

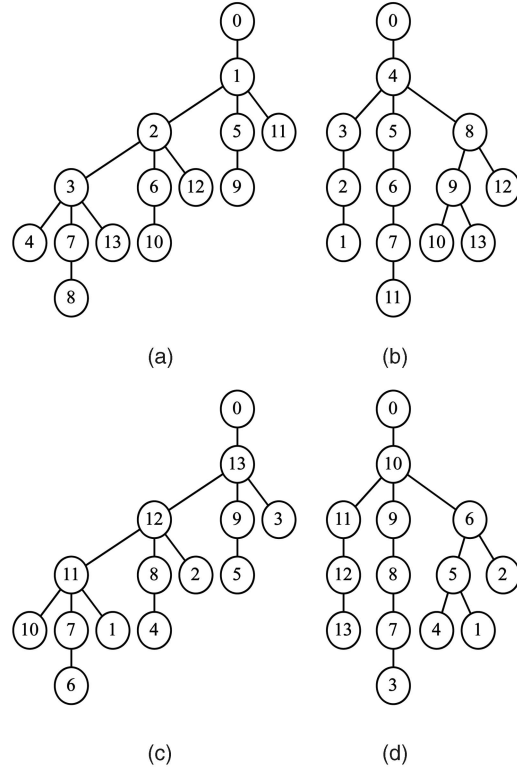
end T_1

Procedure $T_2(N, d)$

begin

1. **for** $x = 1$ **to** $d - 1$ **do**
 $p(x) \leftarrow x + 1$;
2. **for** $x = d$ **to** $N - d$ **do**
if $[x]_d = 0$ **then**
 $p(x) \leftarrow x - d$;
else
 $p(x) \leftarrow x - 1$;
3. **for** $x = N - d + 1$ **to** $N - 1$ **do**
 $p(x) \leftarrow x - d$;

end T_2


 Fig. 2. A set of independent spanning trees on $CR(14, 4)$. (a) T_1 . (b) T_2 . (c) T_3 . (d) T_4 .

To construct T_3 and T_4 , two symmetric procedures with respect to $T_1(N, d)$ and $T_2(N, d)$ are easily obtained by changing vertex x to $N - x$ in the procedures. Iwasaki et al. [9] showed that the resulting spanning trees are independent. For example, based on the above algorithm, four spanning trees rooted at vertex 0 in $CR(14, 4)$ are shown in Fig. 2. Iwasaki et al. used the following notations to represent a certain path: The notation $x \xrightarrow{+i} y$ (respectively, $x \xrightarrow{-i} y$) denotes the path of length 1 from vertex x to vertex y such that $[y - x]_N = i$ (respectively, $[y - x]_N = -i$). Also, the reflexive and transitive closures of $\xrightarrow{+i}$ (respectively, $\xrightarrow{-i}$) are denoted by $\Rightarrow^{+i}$ (respectively, $\Rightarrow^{-i}$). For example, consider T_1 in Fig. 2a. The path $(0, 1, 2, 3, 7, 8)$ connecting root 0 and vertex 8 can be represented by $(0 \Rightarrow^{+1} 1 \xrightarrow{+d} 2 \xrightarrow{+1} 3 \xrightarrow{+1} 7 \xrightarrow{+1} 8)$. In this paper, we follow the use of these notations.

Let $d_G(x, y)$ denote the distance between vertices x and y in a graph G . The height of a spanning tree T rooted at vertex r , denoted by $height(T)$, is the maximum distance from r to any other vertex of T , i.e., $height(T) = \max\{d_T(r, x) | x \neq r\}$. In this case, the path connecting r and x is called a longest path of T . For example, we have $height(T_i) = 5$ for all $i = 1, 2, 3, 4$ in Fig. 2. The path $(0, 1, 2, 3, 7, 8)$ is the longest path in T_1 . To diminish the heights of spanning trees, we shall propose a new algorithm for constructing four independent trees. An observation from Iwasaki et al.'s algorithm reveals that the longest paths of T_1 and T_2 usually have common vertices. A similar circumstance also appears in T_3 and T_4 . For instance, each of the four spanning trees shown in Fig. 2 has a unique longest path, and every longest path contains vertex 7. Thus, if we can redispense the "dead-end" vertices in

$y = 0$. Hence, $d = y + z + z'$. Since $z \neq 0$ and $z' \neq 0$, this implies $d > y + z$ and $d > y + z'$.

Case 4. $d + 1 \leq z + z' < 2d - 1$. Clearly, $z \neq 0$ and $z' \neq 0$. In this case, $\text{remote_node} = (k + k')d - 1 > d - 1$ and $\text{move_column} = [-1 - z - z']_d = 2d - (z + z' + 1)$. Thus, $0 < \text{move_column} < d - 1$. From (*), we have $y = 2d - (z + z' + 1)$. Since $z' \leq d - 1$, it shows that $y + z = 2d - (z + z' + 1) + z = d + [(d - 1) - z'] \geq d$. Contrastingly, since $z \leq d - 1$, it shows that $y + z' = 2d - (z + z' + 1) + z' = d + [(d - 1) - z] \geq d$.

□

By Lemma 2.1 and Lemma 2.3, we summarize the following:

Corollary 2.4. *Let $d \leq x \leq N - d$ such that $[x]_d = z$ and $[N - x]_d = z'$. Then, all of the following statements are true:*

1. If $z = z' = 0$, then $y = 0$.
2. If $z = 0$ and $z' \neq 0$, then $d > y + z'$.
3. If $z \neq 0$ and $z' = 0$, then $d > y + z$.
4. If $z \neq 0$ and $z' \neq 0$, then
 - a. $d > y + z$ and $d > y + z'$ whenever $1 \leq z + z' \leq d$;
 - b. $d \leq y + z$ and $d \leq y + z'$ whenever $d < z + z' < 2d - 1$.

3 REFINED PROCEDURES FOR CASE $[N]_d \neq 0$

We first describe our algorithm to construct four independent spanning trees rooted at vertex 0 in $CR(N, d)$ under the case $[N]_d \neq 0$ and $2 \leq d < N/2$. For distinguishability, the spanning trees generated in this section are denoted by T_1^* , T_2^* , T_3^* , and T_4^* , respectively. The following are the refinements of procedures $T1(N, d)$ and $T2(N, d)$, which can be used to construct T_1^* and T_2^* :

Procedure $T1^*(N, d)$

begin

1. $y = \text{COLUMN_SPAN}(N, d)$;
2. **for** $x = 1$ **to** $d - 1$ **do**
 $p(x) \leftarrow x - 1$;
3. **for** $x = d$ **to** $N - d$ **do**
if $[x]_d = 0$ **then**
 if $y > 0$ and $x/d \leq \lfloor N/2d \rfloor$ **then**
 $p(x) \leftarrow x + 1$;
 else
 $p(x) \leftarrow x - 1$;
else
 if $[N - x]_d = 0$ **then**
 if $y > 0$ and $(N - x)/d \leq \lfloor N/2d \rfloor$ **then**
 $p(x) \leftarrow x + 1$;
 else
 $p(x) \leftarrow x - d$;
 else
 if $d \leq y + [x]_d$ **then** // It guarantees $y > 0$
 $p(x) \leftarrow x + d$;
 else
 $p(x) \leftarrow x - d$;
- 4. **for** $x = N - d + 1$ **to** $N - 1$ **do**

$p(x) \leftarrow [x + d]_N$;

end $T1^*$

Procedure $T2^*(N, d)$

begin

1. $y = \text{COLUMN_SPAN}(N, d)$;
2. **for** $x = 1$ **to** $d - 1$ **do**
 $p(x) \leftarrow x + 1$;
3. **for** $x = d$ **to** $N - d$ **do**
if $[x]_d = 0$ **then**
 $p(x) \leftarrow x - d$;
else
 if $[N - x]_d = 0$ **then**
 if $y > 0$ and $(N - x)/d \leq \lfloor N/2d \rfloor$ **then**
 $p(x) \leftarrow x - d$;
 else
 $p(x) \leftarrow x - 1$;
 else
 if $d \leq y + [x]_d$ **then** // It guarantees $y > 0$
 $p(x) \leftarrow x + 1$;
 else
 $p(x) \leftarrow x - 1$;
- 4. **for** $x = N - d + 1$ **to** $N - 1$ **do**
 $p(x) \leftarrow x - d$;

end $T2^*$

For constructing T_3^* and T_4^* , we use the same scheme as that in Iwasaki et al.'s algorithm to obtain two symmetric procedures that change vertex x to $N - x$ in $T1^*(N, d)$ and $T2^*(N, d)$, respectively. In the following, we consider $i = 1, 2, 3$, or 4 and let T_i^* be the graph with vertex set $\{0, 1, \dots, N - 1\}$ and an edge set specified by one of the above procedures. Obviously, the edge joining x and its parent in T_i^* is the same as that in T_i for $1 \leq x \leq d - 1$ and $N - d + 1 \leq x \leq N - 1$. However, the case is more complicated when $d \leq x \leq N - d$. In fact, if y receives a value 0 in Step 1, then all edges constructed by $T_i^*(N, d)$ are identical to those in $T_i(N, d)$. Also, it is clear that T_i^* is a spanning subgraph of $CR(N, d)$. To show that T_i^* is a tree, it suffices to show that every vertex $x \neq 0$ in T_i^* has a unique path connecting to vertex 0. Note that this path can be constructed by finding $p(x), p(p(x)), \dots$ repeatedly. For example, the graphs T_i^* , $i = 1, 2, 3, 4$, generated by the refined procedures are shown in Fig. 4 for $CR(14, 4)$. Recall that $y = [7 - 14]_4 = 1$ in this instance. Thus, the path connecting $r = 0$ and vertex $x = 8$ in T_1^* can be constructed by the following steps:

- | | |
|-----------------------------|---|
| $p(8) = 8 - 1 = 7$ | (Step 3 with condition $[8]_4 = 0$ and $8/4 > \lfloor 14/(2 \times 4) \rfloor$), |
| $p(7) = 7 + 4 = 11$ | (Step 3 with condition $[7]_4 = [14 - 7]_4 \neq 0$ and $4 \leq 1 + [7]_4$), |
| $p(11) = [11 + 4]_{14} = 1$ | (Step 4), |
| $p(1) = 1 - 1 = 0$ | (Step 2). |

As a result, we can represent this path as $0 \xrightarrow{+1} y \xrightarrow{-d} x - 1 \xrightarrow{+1} x$ using Iwasaki et al.'s notations. Furthermore, we can describe any path from $r = 0$ to a given vertex x in each of the graphs T_i^* , $i = 1, 2, 3, 4$, under this representation. Since we have verified that any of these referring paths

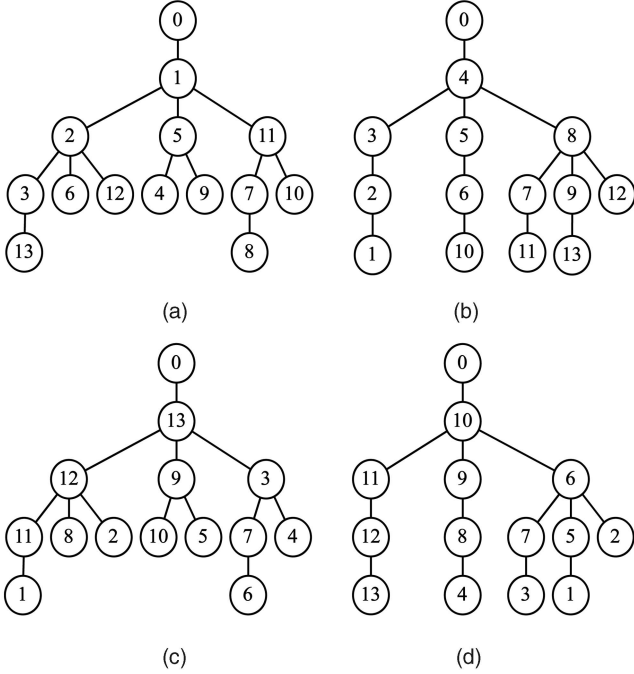


Fig. 4. T_1^* , T_2^* , T_3^* and T_4^* on $CR(14, 4)$. (a) T_1^* . (b) T_2^* . (c) T_3^* . (d) T_4^* .

is the unique path connecting r and x in the graph, this shows that each of the graphs forms a spanning tree. To further reference the unique path from the root to a given vertex x in each spanning tree, we summarize all paths in Appendix A. Moreover, all vertices x within the range between d and $N - d$ are divided into six classes, namely, A , B , C , D , E , and F , respectively. Consequently, we have the following:

Lemma 3.1. T_i^* is a spanning tree of $CR(N, d)$ for $i = 1, 2, 3, 4$.

We are now at a position to show that T_1^* , T_2^* , T_3^* , and T_4^* are independent spanning trees. By symmetry, we only need to prove that the pairs (T_1^*, T_2^*) , (T_1^*, T_3^*) , (T_2^*, T_4^*) , and (T_1^*, T_4^*) are independent. For readability, we merely keep the complete proof for pair (T_1^*, T_2^*) and drop some of tediously algebraic computations for other pairs. All the omitted proofs for showing independence can be achieved by referring to the paths shown in Appendix A in a similar way and are available at <http://www.ntcb.edu.tw/teacher/spade/ChordalRings>. In the following lemma, we assume that x is any vertex of $CR(N, d)$, $y > 0$ is the return value of $COLUMN_SPAN(N, d)$, $z = [x]_d$, and $z' = [N - x]_d$. Recall that $y = 0$ implies $T_i = T_i^*$ for all $i = 1, 2, 3, 4$, and the independence of this case has been proven in [9]. Moreover, by Lemma 2.1, Corollary 2.2, and Corollary 2.4, the assumption $y > 0$ implies $[N]_d \notin \{0, d - 1\}$, $2 < d < \frac{N}{3}$, and either $z \neq 0$ or $z' \neq 0$.

Lemma 3.2. T_1^* and T_2^* are independent spanning trees rooted at 0 of $CR(N, d)$.

Proof. Let P and Q be the unique paths from 0 to x in T_1^* and in T_2^* , respectively. We first consider $1 \leq x \leq d - 1$. In this case, we have $P = (0 \xrightarrow{+1} x)$ and $Q = (0 \xrightarrow{+d} d \xrightarrow{-1} x)$. Then, every internal vertex $a \in P$ satisfies $0 < a < x$,

whereas every internal vertex $b \in Q$ satisfies $x + 1 \leq b \leq d$. Hence, $P \parallel Q$.

Next, we consider $d \leq x \leq N - d$ and $z = 0$. When $x/d \leq \lfloor N/2d \rfloor$ (i.e., $x \in$ class A), these paths are $P = (0 \xrightarrow{+1} 1 \xrightarrow{+d} x + 1 \xrightarrow{-1} x)$ and $Q = (0 \xrightarrow{+d} x)$. Then, every internal vertex $a \in P$ satisfies $[a]_d = 1$, whereas every internal vertex $b \in Q$ satisfies $[b]_d = 0$. This shows that $P \parallel Q$. When $x/d > \lfloor N/2d \rfloor$ (i.e., $x \in$ class B), these paths are $P = (0 \xrightarrow{+1} y \xrightarrow{-d} x - 1 \xrightarrow{+1} x)$ and $Q = (0 \xrightarrow{+d} x)$. Since $0 < y < d - 1$, it is easy to check that every internal vertex $a \in P$ satisfies $[a]_d \neq 0$. Oppositely, we have shown that every internal vertex $b \in Q$ satisfies $[b]_d = 0$. Thus, $P \parallel Q$.

In succession, we consider $d \leq x \leq N - d$ and $z \neq 0$. When $z' = 0$ and $(N - x)/d \leq \lfloor N/2d \rfloor$ (i.e., $x \in$ class C), these paths are $P = (0 \xrightarrow{+1} 1 \xrightarrow{-d} x + 1 \xrightarrow{-1} x)$ and $Q = (0 \xrightarrow{+d} \lfloor (N - d)/2d \rfloor \times d \xrightarrow{+1} \lfloor (N - d)/2d \rfloor \times d + z \xrightarrow{+d} x)$. Then, every internal vertex $a \in P$ and $a \neq 1$ is larger than x , whereas every internal vertex $b \in Q$ is smaller than x . Also, it is easy to see that $1 \notin Q$. This shows that $P \parallel Q$. When $z' = 0$ and $(N - x)/d > \lfloor N/2d \rfloor$ (i.e., $x \in$ class D) or $z' \neq 0$ and $d > y + z$ (i.e., $x \in$ class F), these paths are $P = (0 \xrightarrow{+1} z \xrightarrow{+d} x)$ and $Q = (0 \xrightarrow{+d} x - z \xrightarrow{+1} x)$. The internal vertices of P are $1, 2, \dots, z, z + d, z + 2d, \dots, x - d$, whereas the internal vertices of Q are

$$d, 2d, \dots, x - z, x - z + 1, x - z + 2, \dots, x - 1.$$

Since $z < d$, we have $x - z > x - d$. Thus, we only need to consider the subpaths $P' = (z, z + d, z + 2d, \dots, x - d)$ and $Q' = (d, 2d, \dots, x - z)$ for showing $P \parallel Q$. Obviously, every internal vertex $a \in P'$ satisfies $[a]_d = z \neq 0$, and every internal vertex $b \in Q'$ satisfies $[b]_d = 0$. Thus, $P' \parallel Q'$. When $z' \neq 0$ and $d \leq y + z$ (i.e., $x \in$ class E), these paths are $P = (0 \xrightarrow{+1} [x - N]_d \xrightarrow{-d} x)$ and $Q = (0 \xrightarrow{+d} x + [-x]_d \xrightarrow{-1} x)$. Since $[x - N]_d < d$, every internal vertex $a \in P$ satisfies $1 \leq a < d$ or $x + d \leq a \leq [x - N]_d - d$. Also, $[x]_d = z$ implies $[-x]_d = d - z$. Thus, every internal vertex $b \in Q$ satisfies $d \leq b \leq x + d - z$ or $x + 1 \leq b \leq x + d - z$. Then, either $a < d \leq b$ or $b \leq x + d - z < x + d \leq a$. We conclude $P \parallel Q$.

Finally, we consider $N - d < x \leq N - 1$. When $d \leq y + z$, these paths are $P = (0 \xrightarrow{+1} [x + d]_N \xrightarrow{-d} x)$ and $Q = (0 \xrightarrow{+d} \lfloor (N - d)/d \rfloor \times d \xrightarrow{-1} x - d \xrightarrow{+d} x)$. Then, every internal vertex $a \in P$ satisfies $0 < a \leq [x + d]_N$, whereas every internal vertex $b \in Q$ satisfies $d \leq b \leq \lfloor (N - d)/d \rfloor \times d$. Since $x \leq N - 1$, we have $x + d \leq N + d - 1$ and $[x + d]_N \leq d - 1$. Thus, $a \leq [x + d]_N \leq d - 1 < d \leq b$. We conclude $P \parallel Q$. When $y + z < d$, these paths are $P = (0 \xrightarrow{+1} [x + d]_N \xrightarrow{-d} x)$ and $Q = (0 \xrightarrow{+d} \lfloor (N - d)/d \rfloor \times d \xrightarrow{+1} x - d \xrightarrow{+d} x)$. For any internal vertex $a \in P$ and any internal vertex $b \in Q$, an argument similar to the above case shows that $a \leq [x + d]_N \leq d - 1 < d \leq b$ and it further implies $P \parallel Q$. $\square$

As we have mentioned before that we have verified the pairs (T_1^*, T_3^*) , (T_2^*, T_4^*) , and (T_1^*, T_4^*) for independence, an immediate consequence is the following:

Theorem 1. *For any $2 \leq d < N/2$, T_1^* , T_2^* , T_3^* , and T_4^* are independent spanning trees rooted at 0 of $CR(N, d)$.*

4 REFINED PROCEDURES FOR CASE $[N]_d = 0$

In this section, we design an algorithm to construct four independent spanning trees T_1^+ , T_2^+ , T_3^+ , and T_4^+ rooted at vertex 0 in $CR(N, d)$ under $[N]_d = 0$. For $d = 2$, we suggest the usage of Iwasaki et al.'s algorithm. The following refinements of procedures $T1(N, d)$ and $T2(N, d)$ to construct T_1^+ and T_2^+ are considered for $2 < d < N/2$:

Procedure $T1^+(N, d)$

begin

1. **for** $x = 1$ **to** $d - 1$ **do**
 $p(x) \leftarrow x - 1$;
 2. **for** $x = d$ **to** $N - d$ **do**
if $[x]_d = 0$ **then**
 $p(x) \leftarrow x + 1$;
else
if $[x]_d < \lfloor (d + 1)/2 \rfloor$ **then**
if $\lfloor x/d \rfloor \leq \lfloor N/2d \rfloor$ **then**
 $p(x) \leftarrow x - d$;
else
 $p(x) \leftarrow x + d$;
if $[x]_d = d/2$ and $[d]_2 = 0$ **then**
if $\lfloor x/d \rfloor \leq \lfloor N/2d \rfloor$ **then**
 $p(x) \leftarrow x - d$;
else
 $p(x) \leftarrow x - 1$;
if $[x]_d > \lfloor d/2 \rfloor$ **then**
 $p(x) \leftarrow x - 1$;
 3. **for** $x = N - d + 1$ **to** $N - 1$ **do**
if $[x]_d \leq \lfloor d/2 \rfloor$ **then**
 $p(x) \leftarrow [x + d]_N$;
else
 $p(x) \leftarrow x - 1$;
- end $T1^+$**

Procedure $T2^+(N, d)$

begin

1. **for** $x = 1$ **to** $d - 1$ **do**
if $x \leq \lfloor (d - 1)/2 \rfloor$ **then**
 $p(x) \leftarrow x + d$;
else
 $p(x) \leftarrow x + 1$;
2. **for** $x = d$ **to** $N - d$ **do**
if $[x]_d = 0$ **then**
 $p(x) \leftarrow x - d$;
else
if $[x]_d < \lfloor (d + 1)/2 \rfloor$ **then**
if $\lfloor x/d \rfloor \leq \lfloor N/2d \rfloor$ **then**
 $p(x) \leftarrow x - 1$;
else
 $p(x) \leftarrow x - d$;
if $[x]_d = d/2$ and $[d]_2 = 0$ **then**

if $\lfloor x/d \rfloor \leq \lfloor N/2d \rfloor$ **then**

$p(x) \leftarrow x - 1$;

else

$p(x) \leftarrow x - d$;

if $[x]_d > \lfloor d/2 \rfloor$ **then**

if $\lfloor x/d \rfloor \leq \lfloor (N - 3d)/2d \rfloor$ **then**

$p(x) \leftarrow x + 1$;

else

$p(x) \leftarrow x - d$;

3. **for** $x = N - d + 1$ **to** $N - 1$ **do**

if $[x]_d \leq \lfloor d/2 \rfloor$ **then**

$p(x) \leftarrow x - d$;

else

$p(x) \leftarrow [x + d]_N$;

end $T2^+$

We again use symmetry (i.e., it changes vertex x to $N - x$ in the above two procedures) for constructing T_3^+ and T_4^+ . In light of the rules described in these procedures, every vertex $x \neq 0$ in T_i^+ , $i = 1, 2, 3, 4$, has a unique path connecting to vertex 0. We summarize all unique paths from the root to any given vertex x in each of the four spanning trees in Appendix B and, thus, obtain the following result:

Lemma 4.1. T_i^+ is a spanning tree of $CR(N, d)$ for $i = 1, 2, 3, 4$.

In what follows, we give the proof to show the independence of pair (T_1^+, T_2^+) . We assume that x is any vertex of $CR(N, d)$ and $z = [x]_d$. Under assumptions $2 < d < N/2$ and $[N]_d = 0$, it implies $N = kd$ for some integer $k \geq 3$.

Lemma 4.2. T_1^+ and T_2^+ are independent spanning trees rooted at 0 of $CR(N, d)$.

Proof. Let P and Q be the unique paths from 0 to x in T_1^+ and in T_2^+ , respectively. We first consider $1 \leq x \leq d - 1$. In this case, we have $P = (0 \xrightarrow{+1} x)$ and $Q = (0 \xrightarrow{+d} d \xrightarrow{+1} x + d \xrightarrow{-d} x)$ or $Q = (0 \xrightarrow{+d} d \xrightarrow{-1} x)$. Then, every internal vertex $a \in P$ satisfies $0 < a < x$, whereas every internal vertex $b \in Q$ satisfies $x < d \leq b \leq x + d$ or $x + 1 \leq b \leq d$. Hence, $P \parallel Q$.

Next, we consider $d \leq x \leq N - d$ and $z = 0$ (i.e., $x \in$ class A or B). These paths are $P = (0 \xrightarrow{+1} 1 \xrightarrow{+d} x + 1 \xrightarrow{-1} x)$ or $P = (0 \xrightarrow{+1} 1 \xrightarrow{-d} x + 1 \xrightarrow{-1} x)$ and $Q = (0 \xrightarrow{+d} x)$. Then, every internal vertex $a \in P$ satisfies $[a]_d = 1$, whereas every internal vertex $b \in Q$ satisfies $[b]_d = 0$. This shows that $P \parallel Q$.

In succession, we consider $d \leq x \leq N - d$ and $z \neq 0$. When $z < \lfloor (d + 1)/2 \rfloor$ and $\lfloor x/d \rfloor \leq \lfloor N/2d \rfloor$ (i.e., $x \in$ class C or D), these paths are $P = (0 \xrightarrow{+1} z \xrightarrow{+d} x)$ and $Q = (0 \xrightarrow{+d} x - z \xrightarrow{+1} x)$. The internal vertices of P are $1, 2, \dots, z, z + d, z + 2d, \dots, x - d$, whereas the internal vertices of Q are $d, 2d, \dots, x - z, x - z + 1, x - z + 2, \dots, x - 1$. Since $z < d$, we have $x - z > x - d$. Thus, we only need to consider the subpaths $P' = (z, z + d, z + 2d, \dots, x - d)$ and $Q' = (d, 2d, \dots, x - z - d)$ for showing $P \parallel Q$. Obviously, every internal vertex $a \in P'$ satisfies

$[a]_d = z \neq 0$, and every internal vertex $b \in Q'$ satisfies $[b]_d = 0$. Thus, $P' \parallel Q'$. When $z < \lfloor (d+1)/2 \rfloor$ and $\lfloor x/d \rfloor > \lfloor N/2d \rfloor$ (i.e., $x \in \text{class } E$), these paths are $P = (0 \xrightarrow{+1} z \xrightarrow{-d} x)$ and $Q = (0 \xrightarrow{+d} \lfloor N/2d \rfloor \times d \xrightarrow{+1} \lfloor N/2d \rfloor \times d + z \xrightarrow{+d} x)$. The internal vertices of P are

$$1, 2, \dots, z, N-d+z, N-2d+z, \dots, x+d,$$

whereas the internal vertices of Q are

$$\begin{aligned} d, 2d, \dots, \lfloor N/2d \rfloor \times d, \lfloor N/2d \rfloor \times d + 1, \lfloor N/2d \rfloor \\ \times d + 2, \dots, \lfloor N/2d \rfloor \times d + z, (\lfloor N/2d \rfloor + 1) \\ \times d + z, (\lfloor N/2d \rfloor + 2) \times d + z, \dots, x-d. \end{aligned}$$

Since $z < d$, we only need to consider the subpaths $P' = (N-d+z, N-2d+z, \dots, x+d)$. Then, every two internal vertices $a \in P'$ and $b \in Q$ satisfy

$$d \leq b \leq x-d < x+d \leq a \leq N-d+z.$$

This shows that $P' \parallel Q$ and, thus, $P \parallel Q$. In the following three classes, we consider $z = d/2$ and $[d]_2 = 0$. When $\lfloor x/d \rfloor \leq \lfloor N/2d \rfloor$ (i.e., $x \in \text{class } F$ or G), these paths are $P = (0 \xrightarrow{+1} z \xrightarrow{+d} x)$ and $Q = (0 \xrightarrow{+d} x-z \xrightarrow{+1} x)$. The internal vertices of P are $1, 2, \dots, z, 3z, 5z, \dots, x-d$, whereas the internal vertices of Q are $d, 2d, \dots, x-z, x-z+1, x-z+2, \dots, x-1$. Since $z < d$, we have $x-d < x-z$. Thus, we only need to consider the subpaths $P' = (3z, 5z, \dots, x-d)$ and $Q' = (d, 2d, \dots, x-3z)$ for showing $P \parallel Q$. Obviously, every internal vertex $a \in P'$ satisfies $[a]_d = z \neq 0$, and every internal vertex $b \in Q'$ satisfies $[b]_d = 0$. Thus, $P' \parallel Q'$. When $\lfloor x/d \rfloor > \lfloor N/2d \rfloor$ (i.e., $x \in \text{class } H$), these paths are $P = (0 \xrightarrow{+1} z-1 \xrightarrow{-d} x-1 \xrightarrow{+1} x)$ and

$$Q = (0 \xrightarrow{+d} \lfloor N/2d \rfloor \times d \xrightarrow{+1} \lfloor N/2d \rfloor \times d + z \xrightarrow{+d} x).$$

The internal vertices of P are

$$1, 2, \dots, z-1, N-d+z-1, N-2d+z-1, \dots, x-1,$$

whereas the internal vertices of Q are

$$\begin{aligned} d, 2d, \dots, \lfloor N/2d \rfloor \times d, \lfloor N/2d \rfloor \times d + 1, \lfloor N/2d \rfloor \\ \times d + 2, \dots, \lfloor N/2d \rfloor \times d + z, (\lfloor N/2d \rfloor + 1) \\ \times d + z, (\lfloor N/2d \rfloor + 2) \times d + z, \dots, x-d. \end{aligned}$$

Since $z-1 < d$, we only need to consider the subpaths $P' = (N-d+z-1, N-2d+z-1, \dots, x-1)$. Then, every two internal vertices $a \in P'$ and $b \in Q$ satisfy $d \leq b \leq x-d < x-1 \leq a \leq N-d+z-1$. This shows that $P' \parallel Q$ and, thus, $P \parallel Q$. When $z > \lfloor d/2 \rfloor$ (i.e., $x \in \text{class } I, J$ or K), these paths are $P = (0 \xrightarrow{+1} \lfloor d/2 \rfloor \xrightarrow{+d} x-z+\lfloor d/2 \rfloor \xrightarrow{+1} x)$ or $P = (0 \xrightarrow{+1} \lfloor (d-1)/2 \rfloor \xrightarrow{-d} x-z+\lfloor (d-1)/2 \rfloor \xrightarrow{+1} x)$ and $Q = (0 \xrightarrow{+d} x+d-z \xrightarrow{-1} x)$ or

$Q = (0 \xrightarrow{+d} (\lfloor N/2d \rfloor - 1) \times d \xrightarrow{-1} (\lfloor N/2d \rfloor - 1) \times d - d + z \xrightarrow{+d} x)$. Obviously, every internal vertex $a \in P$ satisfies $0 < [a]_d < z$, and every internal vertex $b \in Q$ satisfies $[b]_d \geq z$ or $[b]_d = 0$. Thus, $P \parallel Q$.

Finally, we consider $N-d < x \leq N-1$. When $z \leq \lfloor d/2 \rfloor$, these paths are $P = (0 \xrightarrow{+1} z \xrightarrow{-d} x)$ and $Q = (0 \xrightarrow{+d} \lfloor N/2d \rfloor \times d \xrightarrow{+1} \lfloor N/2d \rfloor \times d + z \xrightarrow{+d} x)$. Then, every two internal vertices $a \in P$ and $b \in Q$ satisfy $0 < a \leq z < d \leq b$. Thus, $P \parallel Q$. When $z > \lfloor d/2 \rfloor$, these paths are $P = (0 \xrightarrow{+1} \lfloor d/2 \rfloor \xrightarrow{-d} N-d+\lfloor d/2 \rfloor \xrightarrow{+1} x)$ and $Q = (0 \xrightarrow{+d} d \xrightarrow{-1} z \xrightarrow{-d} x)$. Obviously, every internal vertex $a \in P$ satisfies $0 < [a]_d < z$, and every internal vertex $b \in Q$ satisfies $[b]_d \geq z$ or $[b]_d = 0$. Thus, $P \parallel Q$. $\square$

All the remaining proofs for showing the independence of pairs (T_1^+, T_3^+) , (T_2^+, T_4^+) , and (T_1^+, T_4^+) can be done in a similar way to the proof in Lemma 4.2 and are available at the Web site mentioned previously. Thus, we immediately obtain the following:

Theorem 2. For any $2 < d < N/2$, T_1^+ , T_2^+ , T_3^+ , and T_4^+ are independent spanning trees rooted at 0 of $CR(N, d)$.

5 ANALYSIS OF IMPROVEMENTS

In this section, we compare the results of our algorithm with Iwasaki et al.'s algorithm. The height of spanning trees constructed by Iwasaki et al.'s algorithm can be expressed in terms of the formula $\text{height}(T_i) = d + \lfloor (N-2d)/d \rfloor$, where $i = 1, 2, 3, 4$. To calculate the height of spanning trees established by our algorithm, we need the following notations. Referring to the paths listed in Appendices A and B, we can determine the farthest vertex from the root for each class of vertices. We denote by $f_i(X)$ the vertex in class X that has the farthest distance to $r = 0$ in T_i^* or T_i^+ . Also, we shortly write $d_i(X)$ as the distance between r and $f_i(X)$. Hence, the height of T_i^* (respectively, T_i^+) can be obtained by comparing $d_i(X)$ for every class X .

The analysis of improvements is divided into two cases: $[N]_d \neq 0$ and $[N]_d = 0$. We first consider $[N]_d \neq 0$ as follows: Taking a look at our algorithm, $y = \text{COLUMN_SPAN}(N, d)$ plays an important role to determine the height of independent spanning trees in this case. Since, if $y = 0$, then T_i and T_i^* are identical for all $i = 1, 2, 3, 4$, the analysis considered here only includes $y > 0$. Also, by symmetry, we solely consider the heights of T_1^* and T_2^* . The following two lemmas provide the information about the vertices with the farthest distance from the root and their corresponding distance in the spanning trees T_1^* and T_2^* for each class of vertices between d and $N-d$. For the correctness of the proofs of lemmas appearing in this section, the reader can refer to <http://www.ntcb.edu.tw/teacher/spade/ChordalRings>.

Lemma 5.1. *If $[N]_d \neq 0$ and $y = d - 1 - [N]_d > 0$, then*

$$\begin{cases} f_1(A) = \lfloor \frac{N}{2d} \rfloor \times d & \text{and } d_1(A) = 2 + \lfloor \frac{N}{2d} \rfloor, \\ f_1(B) = (\lfloor \frac{N}{2d} \rfloor + 1) \times d & \text{and } d_1(B) = y + 2 + \lfloor \frac{N-d}{2d} \rfloor, \\ f_1(C) = N - \lfloor \frac{N}{2d} \rfloor \times d & \text{and } d_1(C) = 2 + \lfloor \frac{N}{2d} \rfloor, \\ f_1(D) = N - (\lfloor \frac{N}{2d} \rfloor + 1) \times d & \text{and } d_1(D) = d - y - 1 + \lfloor \frac{N-d}{2d} \rfloor, \\ f_1(E) = 2d - 1 & \text{and } d_1(E) = y + 1 + \lfloor \frac{N-2d}{d} \rfloor, \\ f_1(F) = N - 1 - d & \text{and } d_1(F) = d - y - 1 + \lfloor \frac{N-2d}{d} \rfloor. \end{cases}$$

For convenience, we define two additional classes of vertices $E' = \{x : N - d < x \leq N - 1 \text{ and } d \leq y + [x]_d\}$ and $F' = \{N - d < x \leq N - 1 \text{ and } d > y + [x]_d\}$ in the following lemma:

Lemma 5.2. *If $[N]_d \neq 0$ and $y = d - 1 - [N]_d > 0$, then*

$$\begin{cases} f_2(A) = f_2(B) = \lfloor \frac{N-d}{d} \rfloor \times d & \text{and } d_2(A) = d_2(B) = \lfloor \frac{N-d}{d} \rfloor, \\ f_2(C) = N - d & \text{and } d_2(C) = d - y + \lfloor \frac{N-2d}{d} \rfloor, \\ f_2(D) = N - (\lfloor \frac{N}{2d} \rfloor + 1) \times d & \text{and } d_2(D) = d - y - 1 + \lfloor \frac{N-d}{2d} \rfloor, \\ f_2(E) = N + 1 - 2d & \text{and } d_2(E) = y + \lfloor \frac{N-d}{d} \rfloor, \\ f_2(F) = N - 1 - d & \text{and } d_2(F) = d - y - 1 + \lfloor \frac{N-2d}{d} \rfloor, \\ f_2(E') = N + 1 - d & \text{and } d_2(E') = y + 2 + \lfloor \frac{N-2d}{d} \rfloor, \\ f_2(F') = N - 1 & \text{and } d_2(F') = d - y + \lfloor \frac{N-2d}{d} \rfloor. \end{cases}$$

We now calculate the heights of T_1^* and T_2^* as follows:

Theorem 3. *Suppose $[N]_d \neq 0$ and let $w = \lfloor (N - 2d)/d \rfloor$. If $y = 0$, then $\text{height}(T_1^*) = d + w$; otherwise,*

$$\text{height}(T_1^*) = \begin{cases} \max\{d, y + 2 + w\} & \text{if } y \geq \lfloor \frac{d}{2} \rfloor \text{ and } \lfloor \frac{N}{2d} \rfloor = 1, \\ \max\{d, y + 1 + w\} & \text{if } y \geq \lfloor \frac{d}{2} \rfloor \text{ and } \lfloor \frac{N}{2d} \rfloor > 1, \\ \max\{d, d - y - 1 + w\} & \text{if } y \leq \lfloor \frac{d}{2} \rfloor - 1. \end{cases}$$

Proof. In the following proof, we only consider the case of $y > 0$, which implies that $N > 3d$, $[N]_d \notin \{0, d - 1\}$, and $y = d - 1 - [N]_d$ by Lemma 2.1. An easy observation shows that, if $X = \{x : 1 \leq x \leq d - 1\}$, then $f_1(X) = d - 1$ and $d_1(X) = d - 1$. Also, if $X = \{x : N - d + 1 \leq x \leq N - 1\}$, then $f_1(X) = N - 1$ and $d_1(X) = d$. We conclude that only if there is a vertex x with $d \leq x \leq N - d$ such that $d_{T_1^*}(0, x) > d$; otherwise $\text{height}(T_1^*) = d$.

Since $y > 0$, it implies $y + \lfloor (N - d)/2d \rfloor \geq \lfloor N/2d \rfloor$. By Lemma 5.1, we have $d_1(A) = d_1(C) \leq d_1(B)$. Also, under the condition $N > 3d$, it is obvious that

$$\begin{aligned} d_1(D) &= d - y - 1 + \left\lfloor \frac{N - d}{2d} \right\rfloor \leq d - y - 1 + \left\lfloor \frac{N - 2d}{d} \right\rfloor \\ &= d_1(F). \end{aligned}$$

Moreover, if $y \geq \lfloor d/2 \rfloor$, then

$$d - y - 1 \leq d - \lfloor d/2 \rfloor - 1 < d - \lfloor d/2 \rfloor \leq \lfloor d/2 \rfloor + 1 \leq y + 1,$$

which implies $d_1(F) < d_1(E)$. On the other hand, if $y \leq \lfloor d/2 \rfloor - 1$, then $y + 1 \leq d - y - 1$, which implies $d_1(E) \leq d_1(F)$. Therefore, we only need to compare

the distances $d_1(B)$, $d_1(E)$, and $d_1(F)$ for determining the height of T_1^* . Consider the following cases:

Case 1. $3d + 1 \leq N \leq 4d - 2$. Note that $\lfloor (N - d)/2d \rfloor = \lfloor (N - 2d)/d \rfloor = 1$ in this case. By Lemma 5.1, $d_1(B) = y + 3 > d_1(E) = y + 2$. Thus, if $y \geq \lfloor d/2 \rfloor$, then $d_1(B) > d_1(E) > d_1(F)$. In particular, if $[N]_d \leq 2$, then

$$\text{height}(T_1^*) = d_1(B) = y + 3 = (d - 1 - [N]_d) + 3 \geq d;$$

otherwise, $\text{height}(T_1^*) = d$. On the other hand, if $y \leq \lfloor d/2 \rfloor - 1$, then we have $d_1(B) = y + 3 \leq \lfloor d/2 \rfloor + 2 \leq d$ and $d_1(F) = d - y \leq d - (\lfloor d/2 \rfloor - 1) \leq d$ since $d > 2$ by Corollary 2.2. Thus, $\text{height}(T_1^*) = d$ in this case.

Case 2. $4d < N$ and $[N]_d \notin \{0, d - 1\}$. Note that $\lfloor (N - d)/2d \rfloor < \lfloor (N - 2d)/d \rfloor$ in this case. By Lemma 5.1,

$$\begin{aligned} d_1(B) &= y + 2 + \lfloor (N - d)/2d \rfloor \leq y + 1 + \lfloor (N - 2d)/d \rfloor \\ &= d_1(E). \end{aligned}$$

Thus, if $y \geq \lfloor d/2 \rfloor$, then $d_1(E) \geq d_1(B)$ and $d_1(E) > d_1(F)$. In particular, if $\lfloor (N - 2d)/d \rfloor \geq [N]_d$, then

$$\begin{aligned} \text{height}(T_1^*) &= d_1(E) = y + 1 + \lfloor (N - 2d)/d \rfloor \\ &= (d - 1 - [N]_d) + 1 + \lfloor (N - 2d)/d \rfloor \geq d; \end{aligned}$$

otherwise, $\text{height}(T_1^*) = d$. On the other hand, if $y \leq \lfloor d/2 \rfloor - 1$, then $d_1(F) \geq d_1(E) \geq d_1(B)$. In particular, if $\lfloor (N - 2d)/d \rfloor \geq d - [N]_d$, then

$$\begin{aligned} \text{height}(T_1^*) &= d_1(F) = d - y - 1 + \lfloor (N - 2d)/d \rfloor \\ &\geq d - (d - 1 - [N]_d) - 1 + (d - [N]_d) = d; \end{aligned}$$

otherwise, $\text{height}(T_1^*) = d$. $\square$

Theorem 4. *Suppose $[N]_d \neq 0$ and let $w = \lfloor (N - 2d)/d \rfloor$. If $y = 0$, then $\text{height}(T_2^*) = d + w$; otherwise,*

$$\text{height}(T_2^*) = \begin{cases} \max\{d, y + 2 + w\} & \text{if } y \geq \lfloor \frac{d}{2} \rfloor, \\ \max\{d, d - y + w\} & \text{if } y \leq \lfloor \frac{d}{2} \rfloor - 1. \end{cases}$$

Proof. We only consider the case where $y > 0$, which implies that $N > 3d$, $[N]_d \notin \{0, d - 1\}$, and $y = d - 1 - [N]_d$ by Lemma 2.1. An easy observation shows that, if $X = \{x : 1 \leq x \leq d - 1\}$, then $f_2(X) = 1$ and $d_2(X) = d$. We conclude that $\text{height}(T_2^*) = d$ unless there is a vertex x with $d \leq x \leq N - 1$ and $d_{T_2^*}(0, x) > d$.

Since $N > 3d$, it implies $\lfloor (N - d)/d \rfloor < 2 + \lfloor (N - 2d)/d \rfloor$ and $\lfloor (N - d)/2d \rfloor \leq \lfloor (N - 2d)/d \rfloor$. By Lemma 5.1, we have $d_2(A) < d_2(E) < d_2(E')$ and $d_2(D) \leq d_2(F) < d_2(C) = d_2(F')$. Therefore, we only need to compare the distances $d_2(E')$ and $d_2(F')$ for determining the height of T_2^* . From Lemma 5.1, $d_2(E') = y + 2 + \lfloor (N - 2d)/d \rfloor$ and $d_2(F') = d - y + \lfloor (N - 2d)/d \rfloor$.

Case 1. $y \geq \lfloor d/2 \rfloor$. Then, $d - y \leq d - \lfloor d/2 \rfloor \leq \lfloor d/2 \rfloor + 1 < y + 2$. Thus, $d_2(E') > d_2(F')$. In particular, if $\lfloor (N - 2d)/d \rfloor \geq [N]_d - 1$, then

$$\begin{aligned} \text{height}(T_2^*) &= d_2(E') = y + 2 + \lfloor (N - 2d)/d \rfloor \\ &\geq (d - 1 - [N]_d) + 2 + ([N]_d - 1) = d; \end{aligned}$$

otherwise, $\text{height}(T_2^*) = d$.

TABLE 1
The Unique Path in T_1^*

Range of x	Additional conditions			Class	The path from 0 to x
$1 \leq x \leq d-1$	-			-	$0 \xrightarrow{+1} x$
$d \leq x \leq N-d$	$[x]_d = 0$	$x/d \leq \lfloor N/2d \rfloor$		A	$0 \xrightarrow{+1} 1 \xrightarrow{+d} x+1 \xrightarrow{-1} x$
		$x/d > \lfloor N/2d \rfloor$		B	$0 \xrightarrow{+1} y \xrightarrow{-d} x-1 \xrightarrow{+1} x$
	$[x]_d = z \neq 0$	$[N-x]_d = 0$	$(N-x)/d \leq \lfloor N/2d \rfloor$	C	$0 \xrightarrow{+1} 1 \xrightarrow{-d} x+1 \xrightarrow{-1} x$
			$(N-x)/d > \lfloor N/2d \rfloor$	D	$0 \xrightarrow{+1} z \xrightarrow{+d} x$
		$[N-x]_d \neq 0$	$d \leq y+z$	E	$0 \xrightarrow{+1} [x-N]_d \xrightarrow{-d} x$
			$d > y+z$	F	$0 \xrightarrow{+1} z \xrightarrow{+d} x$
$N-d < x \leq N-1$	-			-	$0 \xrightarrow{+1} [x+d]_N \xrightarrow{-d} x$

TABLE 2
The Unique Path in T_2^*

Range of x	Additional conditions			Class	The path from 0 to x
$1 \leq x \leq d-1$	-			-	$0 \xrightarrow{+d} d \xrightarrow{-1} x$
$d \leq x \leq N-d$	$[x]_d = 0$			A, B	$0 \xrightarrow{+d} x$
	$[x]_d = z \neq 0$	$[N-x]_d = 0$	$(N-x)/d \leq \lfloor N/2d \rfloor$	C	$0 \xrightarrow{+d} \lfloor (N-d)/2d \rfloor \times d \xrightarrow{+1} \lfloor (N-d)/2d \rfloor \times d + z \xrightarrow{+d} x$
			$(N-x)/d > \lfloor N/2d \rfloor$	D	$0 \xrightarrow{+d} x - z \xrightarrow{+1} x$
		$[N-x]_d \neq 0$	$d \leq y + z$	E	$0 \xrightarrow{+d} x + [-x]_d \xrightarrow{-1} x$
			$d > y + z$	F	$0 \xrightarrow{+d} x - z \xrightarrow{+1} x$
	$N-d < x \leq N-1$	$d \leq y + [x]_d$			-
$d > y + [x]_d$			-	$0 \xrightarrow{+d} \lfloor (N-d)/d \rfloor \times d \xrightarrow{+1} x - d \xrightarrow{+d} x$	

TABLE 3
The Unique Path in T_3^*

Range of x	Additional conditions			Class	The path from 0 to x
$1 \leq x \leq d-1$	-			-	$0 \xrightarrow{-1} [x-d]_N \xrightarrow{+d} x$
$d \leq x \leq N-d$	$[N-x]_d = 0$	$(N-x)/d \leq \lfloor N/2d \rfloor$		C	$0 \xrightarrow{-1} N-1 \xrightarrow{-d} x-1 \xrightarrow{+1} x$
		$(N-x)/d > \lfloor N/2d \rfloor$		D	$0 \xrightarrow{-1} N-y \xrightarrow{+d} x+1 \xrightarrow{-1} x$
	$[N-x]_d = z' \neq 0$	$[x]_d = 0$	$x/d \leq \lfloor N/2d \rfloor$	A	$0 \xrightarrow{-1} N-1 \xrightarrow{+d} x-1 \xrightarrow{+1} x$
			$x/d > \lfloor N/2d \rfloor$	B	$0 \xrightarrow{-1} N-z' \xrightarrow{-d} x$
		$[x]_d \neq 0$	$d \leq y+z'$	E	$0 \xrightarrow{-1} N-[x]_d \xrightarrow{+d} x$
			$d > y+z'$	F	$0 \xrightarrow{-1} N-z' \xrightarrow{-d} x$
$N-d < x \leq N-1$	-			-	$0 \xrightarrow{-1} x$

TABLE 4
The Unique Path in T_4^*

Range of x	Additional conditions			Class	The path from 0 to x
$1 \leq x \leq d-1$	$d \leq y + \lfloor N-x \rfloor_d$			-	$0 \xrightarrow{-d} N - (\lfloor (N-d)/d \rfloor \times d) \xrightarrow{+1} x + d \xrightarrow{-d} x$
	$d > y + \lfloor N-x \rfloor_d$			-	$0 \xrightarrow{-d} N - (\lfloor (N-d)/d \rfloor \times d) \xrightarrow{-1} x + d \xrightarrow{-d} x$
$d \leq x \leq N-d$	$\lfloor N-x \rfloor_d = 0$			C, D	$0 \xrightarrow{-d} x$
	$\lfloor N-x \rfloor_d = z' \neq 0$	$\lfloor x \rfloor_d = 0$	$x/d \leq \lfloor N/2d \rfloor$	A	$0 \xrightarrow{-d} N - (\lfloor (N-d)/2d \rfloor \times d) \xrightarrow{-1} N - (\lfloor (N-d)/2d \rfloor \times d + z') \xrightarrow{-d} x$
			$x/d > \lfloor N/2d \rfloor$	B	$0 \xrightarrow{-d} x + z' \xrightarrow{-1} x$
		$\lfloor x \rfloor_d \neq 0$	$d \leq y + z'$	E	$0 \xrightarrow{-d} x - \lfloor x - N \rfloor_d \xrightarrow{+1} x$
			$d > y + z'$	F	$0 \xrightarrow{-d} x + z' \xrightarrow{-1} x$
	$N-d < x \leq N-1$	-			-

TABLE 5
 The Unique Path in T_1^+

Range of x	Additional conditions			Class	The path from 0 to x
$1 \leq x \leq d-1$	-			-	$0 \xrightarrow{+1} x$
$d \leq x \leq N-d$	$[x]_d = 0$	$x/d \leq \lfloor N/2d \rfloor$		A	$0 \xrightarrow{+1} 1 \xrightarrow{+d} x+1 \xrightarrow{-1} x$
		$x/d > \lfloor N/2d \rfloor$		B	$0 \xrightarrow{+1} 1 \xrightarrow{-d} x+1 \xrightarrow{-1} x$
	$[x]_d = z \neq 0$	$z < \lfloor \frac{d+1}{2} \rfloor$	$\lfloor x/d \rfloor \leq \lfloor (N-3d)/2d \rfloor$	C	$0 \xrightarrow{+1} z \xrightarrow{+d} x$
			$\lfloor (N-3d)/2d \rfloor < \lfloor x/d \rfloor \leq \lfloor N/2d \rfloor$	D	
			$\lfloor x/d \rfloor > \lfloor N/2d \rfloor$	E	$0 \xrightarrow{+1} z \xrightarrow{-d} x$
			$\lfloor x/d \rfloor \leq \lfloor (N-3d)/2d \rfloor$	F	$0 \xrightarrow{+1} z \xrightarrow{+d} x$
		$z = d/2$	$\lfloor (N-3d)/2d \rfloor < \lfloor x/d \rfloor \leq \lfloor N/2d \rfloor$	G	
			$\lfloor x/d \rfloor > \lfloor N/2d \rfloor$	H	$0 \xrightarrow{+1} z-1 \xrightarrow{-d} x-1 \xrightarrow{+1} x$
		$z > \lfloor d/2 \rfloor$	$\lfloor x/d \rfloor \leq \lfloor (N-3d)/2d \rfloor$	I	$0 \xrightarrow{+1} \lfloor d/2 \rfloor \xrightarrow{+d} x-z+\lfloor d/2 \rfloor$
			$\lfloor (N-3d)/2d \rfloor < \lfloor x/d \rfloor \leq \lfloor N/2d \rfloor$	J	$\xrightarrow{+1} x$
			$\lfloor x/d \rfloor > \lfloor N/2d \rfloor$	K	$0 \xrightarrow{+1} \lfloor (d-1)/2 \rfloor \xrightarrow{-d} x-z+\lfloor (d-1)/2 \rfloor \xrightarrow{+1} x$
$N-d < x \leq N-1$	$[x]_d \leq \lfloor d/2 \rfloor$			-	$0 \xrightarrow{+1} z \xrightarrow{-d} x$
	$[x]_d > \lfloor d/2 \rfloor$			-	$0 \xrightarrow{+1} \lfloor d/2 \rfloor \xrightarrow{-d} N-d+\lfloor d/2 \rfloor \xrightarrow{+1} x$

 TABLE 6
 The Unique Path in T_2^+

Range of x	Additional conditions			Class	The path from 0 to x	
$1 \leq x \leq d-1$	$[x]_d \leq \lfloor (d-1)/2 \rfloor$			-	$0 \xrightarrow{+d} d \xrightarrow{+1} x+d \xrightarrow{-d} x$	
	$[x]_d > \lfloor (d-1)/2 \rfloor$			-	$0 \xrightarrow{+d} d \xrightarrow{-1} x$	
$d \leq x \leq N-d$	$[x]_d = 0$	$x/d \leq \lfloor N/2d \rfloor$		A	$0 \xrightarrow{+d} x$	
		$x/d > \lfloor N/2d \rfloor$		B		
	$[x]_d = z \neq 0$	$z < \lfloor \frac{d+1}{2} \rfloor$	$[x/d] \leq \lfloor (N-3d)/2d \rfloor$		C	$0 \xrightarrow{+d} x-z \xrightarrow{+1} x$
			$\lfloor (N-3d)/2d \rfloor < [x/d] \leq \lfloor N/2d \rfloor$		D	
			$[x/d] > \lfloor N/2d \rfloor$		E	$0 \xrightarrow{+d} \lfloor N/2d \rfloor \times d \xrightarrow{+1} \lfloor N/2d \rfloor \times d + z \xrightarrow{+d} x$
		$z = d/2$ $[d]_2 = 0$	$[x/d] \leq \lfloor (N-3d)/2d \rfloor$		F	$0 \xrightarrow{+d} x-z \xrightarrow{+1} x$
			$\lfloor (N-3d)/2d \rfloor < [x/d] \leq \lfloor N/2d \rfloor$		G	
			$[x/d] > \lfloor N/2d \rfloor$		H	$0 \xrightarrow{+d} \lfloor N/2d \rfloor \times d \xrightarrow{+1} \lfloor N/2d \rfloor \times d + z \xrightarrow{+d} x$
		$z > \lfloor d/2 \rfloor$	$[x/d] \leq \lfloor (N-3d)/2d \rfloor$		I	$0 \xrightarrow{+d} x+d-z \xrightarrow{-1} x$
			$\lfloor (N-3d)/2d \rfloor < [x/d] \leq \lfloor N/2d \rfloor$		J	$0 \xrightarrow{+d} (\lfloor N/2d \rfloor - 1) \times d \xrightarrow{-1} (\lfloor N/2d \rfloor - 1) \times d - d + z \xrightarrow{+d} x$
$[x/d] > \lfloor N/2d \rfloor$			K			
$N-d < x \leq N-1$	$[x]_d \leq \lfloor d/2 \rfloor$			-	$0 \xrightarrow{+d} \lfloor N/2d \rfloor \times d \xrightarrow{+1} \lfloor N/2d \rfloor \times d + z \xrightarrow{+d} x$	
	$[x]_d > \lfloor d/2 \rfloor$			-	$0 \xrightarrow{+d} d \xrightarrow{-1} z \xrightarrow{-d} x$	

 TABLE 7
 The Unique Path in T_3^+

Range of x	Additional conditions		Class	The path from 0 to x	
$1 \leq x \leq d-1$	$[x]_d \leq \lfloor (d-1)/2 \rfloor$		-	$0 \xrightarrow{-1} N - \lfloor d/2 \rfloor \xrightarrow{+d} d - \lfloor d/2 \rfloor \xrightarrow{-1} x$	
	$[x]_d > \lfloor (d-1)/2 \rfloor$		-	$0 \xrightarrow{-1} N - d + z \xrightarrow{+d} x$	
$d \leq x \leq N-d$	$[x]_d = 0$	$(N-x)/d \leq \lfloor N/2d \rfloor$	A'	$0 \xrightarrow{-1} N-1 \xrightarrow{-d} x-1 \xrightarrow{+1} x$	
		$(N-x)/d > \lfloor N/2d \rfloor$	B'	$0 \xrightarrow{-1} N-1 \xrightarrow{+d} x-1 \xrightarrow{+1} x$	
	$[x]_d = z \neq 0$	$z < \lfloor \frac{d+1}{2} \rfloor$	$\lfloor x/d \rfloor \leq \lfloor (N-3d)/2d \rfloor$	C	$0 \xrightarrow{-1} N - \lfloor (d-1)/2 \rfloor \xrightarrow{+d} x + d - z - \lfloor (d-1)/2 \rfloor \xrightarrow{-1} x$
			$\lfloor (N-3d)/2d \rfloor < \lfloor x/d \rfloor \leq \lfloor N/2d \rfloor$	D	$0 \xrightarrow{-1} N - \lfloor d/2 \rfloor \xrightarrow{-d} x + d - z - \lfloor d/2 \rfloor \xrightarrow{-1} x$
			$\lfloor x/d \rfloor > \lfloor N/2d \rfloor$	E	$x + d - z - \lfloor d/2 \rfloor \xrightarrow{-1} x$
		$z = d/2$ $[d]_2 = 0$	$\lfloor x/d \rfloor \leq \lfloor (N-3d)/2d \rfloor$	F	$0 \xrightarrow{-1} N - z + 1 \xrightarrow{+d} x + 1 \xrightarrow{-1} x$
			$\lfloor (N-3d)/2d \rfloor < \lfloor x/d \rfloor \leq \lfloor N/2d \rfloor$	G	$0 \xrightarrow{-1} N - z \xrightarrow{-d} x$
			$\lfloor x/d \rfloor > \lfloor N/2d \rfloor$	H	
		$z > \lfloor d/2 \rfloor$	$\lfloor x/d \rfloor \leq \lfloor (N-3d)/2d \rfloor$	I	$0 \xrightarrow{-1} N - d + z \xrightarrow{+d} x$
			$\lfloor (N-3d)/2d \rfloor < \lfloor x/d \rfloor \leq \lfloor N/2d \rfloor$	J	$0 \xrightarrow{-1} N - d + z \xrightarrow{-d} x$
			$\lfloor x/d \rfloor > \lfloor N/2d \rfloor$	K	
		$N-d < x \leq N-1$	$\lfloor x \rfloor_d \leq \lfloor d/2 \rfloor$		-
$\lfloor x \rfloor_d > \lfloor d/2 \rfloor$			-		

TABLE 8
The Unique Path in T_4^+

Range of x	Additional conditions		Class	The path from 0 to x
$1 \leq x \leq d-1$	$[x]_d \leq \lfloor (d-1)/2 \rfloor$		-	$0 \xrightarrow{-d} N-d \xrightarrow{+1} N-d+z \xrightarrow{+d} x$
	$[x]_d > \lfloor (d-1)/2 \rfloor$		-	$0 \xrightarrow{-d} N - \lfloor N/2d \rfloor \times d \xrightarrow{-1} N - \lfloor N/2d \rfloor \times d - d + z \xrightarrow{-d} x$
$d \leq x \leq N-d$	$[x]_d = 0$	$x/d \leq \lfloor N/2d \rfloor$	A	$0 \xrightarrow{-d} x$
		$x/d > \lfloor N/2d \rfloor$	B	
	$z < \lfloor \frac{d+1}{2} \rfloor$	$\lfloor x/d \rfloor \leq \lfloor (N-3d)/2d \rfloor$	C	$0 \xrightarrow{-d} N - (\lceil N/2d \rceil - 1) \times d \xrightarrow{+1}$
		$\lfloor (N-3d)/2d \rfloor < \lfloor x/d \rfloor \leq \lfloor N/2d \rfloor$	D	$N - (\lceil N/2d \rceil - 1) \times d + z \xrightarrow{-d} x$
		$\lfloor x/d \rfloor > \lfloor N/2d \rfloor$	E	$0 \xrightarrow{-d} x - z \xrightarrow{+1} x$
	$z = d/2$ $[d]_2 = 0$	$\lfloor x/d \rfloor \leq \lfloor (N-3d)/2d \rfloor$	F	$0 \xrightarrow{-d} N - \lfloor N/2d \rfloor \times d \xrightarrow{-1}$
		$\lfloor (N-3d)/2d \rfloor < \lfloor x/d \rfloor \leq \lfloor N/2d \rfloor$	G	$N - \lfloor N/2d \rfloor \times d - z \xrightarrow{-d} x$
		$\lfloor x/d \rfloor > \lfloor N/2d \rfloor$	H	$0 \xrightarrow{-d} x + z \xrightarrow{-1} x$
	$z > \lfloor d/2 \rfloor$	$\lfloor x/d \rfloor \leq \lfloor (N-3d)/2d \rfloor$	I	$0 \xrightarrow{-d} N - \lfloor N/2d \rfloor \times d \xrightarrow{-1}$
		$\lfloor (N-3d)/2d \rfloor < \lfloor x/d \rfloor \leq \lfloor N/2d \rfloor$	J	$N - \lfloor N/2d \rfloor \times d - d + z \xrightarrow{-d} x$
$N-d < x \leq N-1$	$[x]_d \leq \lfloor d/2 \rfloor$		-	$0 \xrightarrow{-d} N-d \xrightarrow{+1} x$
	$[x]_d > \lfloor d/2 \rfloor$		-	$0 \xrightarrow{-d} N-d \xrightarrow{-1} x-d \xrightarrow{+d} x$

TABLE 9
The Comparison of Heights of Spanning Trees

d	$i \setminus N$	31	32	33	34	35	36	37	38	39
3	1	10/11	11/11	7/12	11/12	12/12	8/13	12/13	13/13	8/14
	2	11/11	11/11	11/12	12/12	12/12	12/13	13/13	13/13	13/14
4	1	9/9	7/10	9/10	8/10	10/10	7/11	10/11	9/11	11/11
	2	9/9	9/10	10/10	9/10	10/10	10/11	11/11	10/11	11/11
5	1	8/9	7/9	7/9	9/9	7/10	9/10	8/10	8/10	10/10
	2	9/9	8/9	8/9	9/9	8/10	10/10	9/10	9/10	10/10
6	1	8/9	7/9	6/9	7/9	9/9	8/10	9/10	8/10	7/10
	2	9/9	8/9	7/9	8/9	9/9	8/10	10/10	9/10	8/10
7	1	7/9	7/9	7/9	9/9	8/10	9/10	8/10	7/10	7/10
	2	7/9	7/9	8/9	9/9	7/10	10/10	9/10	8/10	8/10
8	1	9/9	9/10	9/10	8/10	8/10	8/10	8/10	8/10	10/10
	2	9/9	7/10	10/10	9/10	8/10	8/10	8/10	9/10	10/10
9	1	9/10	9/10	9/10	9/10	10/10	10/11	10/11	9/11	9/11
	2	9/10	9/10	9/10	9/10	10/10	7/11	11/11	10/11	9/11
10	1	-	-	10/11	10/11	10/11	10/11	10/11	10/11	11/11
	2	-	-	10/11	10/11	10/11	10/11	10/11	10/11	11/11
11	1	-	-	-	-	-	11/12	11/12	11/12	11/12
	2	-	-	-	-	-	11/12	11/12	11/12	11/12
12	1	-	-	-	-	-	-	-	-	12/13
	2	-	-	-	-	-	-	-	-	12/13
13	1	-	-	-	-	-	-	-	-	-
	2	-	-	-	-	-	-	-	-	-
14	1	-	-	-	-	-	-	-	-	-
	2	-	-	-	-	-	-	-	-	-
15	1	-	-	-	-	-	-	-	-	-
	2	-	-	-	-	-	-	-	-	-

Case 2. $y \leq \lfloor d/2 \rfloor - 1$. Then, $y+2 \leq d-y$ and $d_2(E') \leq d_2(F')$. In particular, if

$$\lfloor (N-2d)/d \rfloor \geq d-1 - [N]_d,$$

then

$$\begin{aligned} height(T_2^*) &= d_2(F') = d-y + \lfloor (N-2d)/d \rfloor \\ &\geq d - (d-1 - [N]_d) + (d-1 - [N]_d) = d; \end{aligned}$$

otherwise, $height(T_2^*) = d$. $\square$

Next, we consider the improvement for case $[N]_d = 0$.

Since, if $d=2$, then T_i and T_i^+ are identical for all

TABLE 10
 The Comparison of Heights of Spanning Trees (Continued)

d	$i \setminus N$	40	41	42	43	44	45	46	47	48
3	1	13/14	14/14	9/15	14/15	15/15	9/16	15/16	16/16	10/17
	2	14/14	14/14	14/15	15/15	15/15	15/16	16/16	16/16	16/17
4	1	8/12	11/12	10/12	12/12	8/13	12/13	11/13	13/13	9/14
	2	11/12	12/12	11/12	12/12	12/13	13/13	12/13	13/13	13/14
5	1	8/11	10/11	9/11	9/11	11/11	8/12	11/12	10/12	10/12
	2	9/11	11/11	10/11	10/11	11/11	10/12	12/12	11/12	11/12
6	1	8/10	10/10	8/11	10/11	9/11	8/11	9/11	11/11	9/12
	2	9/10	10/10	9/11	11/11	10/11	9/11	10/11	11/11	10/12
7	1	8/10	10/10	9/11	10/11	9/11	8/11	8/11	9/11	11/11
	2	9/10	10/10	8/11	11/11	10/11	9/11	9/11	10/11	11/11
8	1	9/11	10/11	9/11	8/11	8/11	8/11	9/11	11/11	10/12
	2	8/11	11/11	10/11	9/11	8/11	9/11	10/11	11/11	9/12
9	1	9/11	9/11	9/11	9/11	11/11	10/12	11/12	10/12	9/12
	2	9/11	9/11	9/11	10/11	11/11	8/12	12/12	11/12	10/12
10	1	11/12	11/12	10/12	10/12	10/12	10/12	10/12	10/12	10/12
	2	8/12	12/12	11/12	10/12	10/12	10/12	10/12	10/12	11/12
11	1	11/12	11/12	11/12	12/12	12/13	12/13	11/13	11/13	11/13
	2	11/12	11/12	11/12	12/12	8/13	13/13	12/13	11/13	11/13
12	1	12/13	12/13	12/13	12/13	12/13	12/13	12/13	13/13	13/14
	2	12/13	12/13	12/13	12/13	12/13	12/13	12/13	13/13	9/14
13	1	-	-	13/14	13/14	13/14	13/14	13/14	13/14	13/14
	2	-	-	13/14	13/14	13/14	13/14	13/14	13/14	13/14
14	1	-	-	-	-	-	14/15	14/15	14/15	14/15
	2	-	-	-	-	-	14/15	14/15	14/15	14/15
15	1	-	-	-	-	-	-	-	-	15/16
	2	-	-	-	-	-	-	-	-	15/16

$i = 1, 2, 3, 4$, the analysis considered here only includes $d \geq 3$. Also, by symmetry, we solely consider the heights of T_1^+ and T_2^+ . Similar to Lemma 5.1 and Lemma 5.2, the following two lemmas provide the information about the vertices with the farthest distance from the root and their corresponding distance in the spanning trees T_1^+ and T_2^+ for each class of vertices between d and $N - d$:

Lemma 5.3. If $[N]_d = 0$, then

$$\left\{ \begin{array}{ll} f_1(A) = \lfloor \frac{N}{2d} \rfloor \times d & \text{and } d_1(A) = 2 + \lfloor \frac{N}{2d} \rfloor, \\ f_1(B) = (\lfloor \frac{N}{2d} \rfloor + 1) \times d & \text{and } d_1(B) = 2 + \lfloor \frac{N-d}{2d} \rfloor, \\ f_1(C) = \lfloor \frac{d+1}{2} \rfloor - 1 & \text{and } d_1(C) = \lfloor \frac{d+1}{2} \rfloor \\ \quad + \lfloor \frac{N-3d}{2d} \rfloor \times d & \quad - 1 + \lfloor \frac{N-3d}{2d} \rfloor, \\ f_1(D) = \lfloor \frac{d+1}{2} \rfloor - 1 & \text{and } d_1(D) = \lfloor \frac{d+1}{2} \rfloor - 1 + \lfloor \frac{N}{2d} \rfloor, \\ \quad + \lfloor \frac{N}{2d} \rfloor \times d & \\ f_1(E) = \lfloor \frac{d+1}{2} \rfloor - 1 & \text{and } d_1(E) = \lfloor \frac{d+1}{2} \rfloor \\ \quad + (\lfloor \frac{N}{2d} \rfloor + 1) \times d & \quad - 1 + \lfloor \frac{N-d}{2d} \rfloor, \\ f_1(F) = \frac{d}{2} + \lfloor \frac{N-3d}{2d} \rfloor \times d & \text{and } d_1(F) = \frac{d}{2} + \lfloor \frac{N-3d}{2d} \rfloor, \\ f_1(G) = \frac{d}{2} + \lfloor \frac{N}{2d} \rfloor \times d & \text{and } d_1(G) = \frac{d}{2} + \lfloor \frac{N}{2d} \rfloor, \\ f_1(H) = \frac{d}{2} + (\lfloor \frac{N}{2d} \rfloor + 1) \times d & \text{and } d_1(H) = \frac{d}{2} + \lfloor \frac{N-d}{2d} \rfloor, \\ f_1(I) = d - 1 + \lfloor \frac{N-3d}{2d} \rfloor \times d & \text{and } d_1(I) = d - 1 + \lfloor \frac{N-3d}{2d} \rfloor, \\ f_1(J) = d - 1 + \lfloor \frac{N}{2d} \rfloor \times d & \text{and } d_1(J) = d - 1 + \lfloor \frac{N}{2d} \rfloor, \\ f_1(K) = d - 1 & \text{and } d_1(K) = d - 1 + \lfloor \frac{N-d}{2d} \rfloor. \\ \quad + (\lfloor \frac{N}{2d} \rfloor + 1) \times d & \end{array} \right.$$

For convenience, we define two additional classes of vertices $J' = \{x : N - d < x \leq N - 1 \text{ and } [x]_d \leq \lfloor d/2 \rfloor\}$ and

$K' = \{N - d < x \leq N - 1 \text{ and } [x]_d > \lfloor d/2 \rfloor\}$ in the following lemma:

Lemma 5.4. If $[N]_d = 0$, then

$$\left\{ \begin{array}{ll} f_2(A) = \lfloor \frac{N}{2d} \rfloor \times d & \text{and } d_2(A) = \lfloor \frac{N}{2d} \rfloor, \\ f_2(B) = N - d & \text{and } d_2(B) = \frac{N}{d} - 1, \\ f_2(C) = \lfloor \frac{N-3d}{2d} \rfloor \times d & \text{and } d_2(C) = \lfloor \frac{N-3d}{2d} \rfloor \\ \quad + \lfloor \frac{d+1}{2} \rfloor - 1 & \quad + \lfloor \frac{d+1}{2} \rfloor - 1, \\ f_2(D) = \lfloor \frac{N}{2d} \rfloor \times d & \text{and } d_2(D) = \lfloor \frac{N}{2d} \rfloor \\ \quad + \lfloor \frac{d+1}{2} \rfloor - 1 & \quad + \lfloor \frac{d+1}{2} \rfloor - 1, \\ f_2(E) = N - 2d & \text{and } d_2(E) = \frac{N}{d} - 2 \\ \quad + \lfloor \frac{d+1}{2} \rfloor - 1 & \quad + \lfloor \frac{d+1}{2} \rfloor - 1, \\ f_2(F) = \lfloor \frac{N-3d}{2d} \rfloor \times d + \frac{d}{2} & \text{and } d_2(F) = \lfloor \frac{N-3d}{2d} \rfloor + \frac{d}{2}, \\ f_2(G) = \lfloor \frac{N}{2d} \rfloor \times d + \frac{d}{2} & \text{and } d_2(G) = \lfloor \frac{N}{2d} \rfloor + \frac{d}{2}, \\ f_2(H) = N - 2d + \frac{d}{2} & \text{and } d_2(H) = \frac{N}{d} - 2 + \frac{d}{2}, \\ f_2(I) = \lfloor \frac{N-3d}{2d} \rfloor \times d + \lfloor \frac{d}{2} \rfloor + 1 & \text{and } d_2(I) = \lfloor \frac{N-3d}{2d} \rfloor + d - \lfloor \frac{d}{2} \rfloor, \\ f_2(J) = \lfloor \frac{N}{2d} \rfloor \times d + \lfloor \frac{d}{2} \rfloor + 1 & \text{and } d_2(J) = \lfloor \frac{N}{2d} \rfloor + d - \lfloor \frac{d}{2} \rfloor, \\ f_2(K) = N - 2d + \lfloor \frac{d}{2} \rfloor + 1 & \text{and } d_2(K) = \frac{N}{d} - 2 + d - \lfloor \frac{d}{2} \rfloor, \\ f_2(J') = N - d + \lfloor \frac{d}{2} \rfloor & \text{and } d_2(J') = \frac{N}{d} - 1 + \lfloor \frac{d}{2} \rfloor, \\ f_2(K') = N - d + 1 + \lfloor \frac{d}{2} \rfloor & \text{and } d_2(K') = d - \lfloor \frac{d}{2} \rfloor + 1. \end{array} \right.$$

We now calculate the heights of T_1^+ and T_2^+ as follows:

Theorem 5. If $d \geq 3$, then $\text{height}(T_1^+) = d - 1 + \lfloor N/2d \rfloor$.

Proof. An easy observation shows that, if

$$X = \{x : 1 \leq x \leq d - 1\},$$

then $f_1(X) = d - 1$ and $d_1(X) = d - 1$. Also, if $X = \{x : N - d + 1 \leq x \leq N - 1\}$, then $f_1(X) = N - 1$ and $d_1(X) = d$. We now show that there is a vertex x with $d \leq x \leq N - d$ such that $d_{T_1^+}(0, x) \geq d$.

Since $N = kd$ for some integer $k \geq 3$, by Lemma 5.3, we have $d_1(B) \leq d_1(A)$, $d_1(C) < d_1(E) \leq d_1(D)$, $d_1(F) < d_1(H) \leq d_1(G)$, and $d_1(I) < d_1(K) \leq d_1(J)$. Thus, we compare the distances $d_1(A)$, $d_1(D)$, $d_1(G)$, and $d_1(J)$ to find the maximum. Since $d \geq 3$, we obtain $d_1(D) < d_1(G) < d_1(J)$ and $d_1(A) \leq d_1(J)$. Therefore, $height(T_1^+) = d_1(J) = d - 1 + \lfloor N/2d \rfloor$. $\square$

Theorem 6. If $d \geq 3$, then $height(T_2^+) = N/d - 1 + \lfloor d/2 \rfloor$.

Proof. An easy observation shows that, if

$$X = \{x : 1 \leq x \leq d - 1\},$$

then $f_2(X) = \lfloor (d - 1)/2 \rfloor$ and $d_2(X) = 2 + \lfloor (d - 1)/2 \rfloor$. We now show that there is a vertex x with $d \leq x \leq N - 1$ such that $d_{T_2^+}(0, x) \geq 2 + \lfloor (d - 1)/2 \rfloor$.

Since $N = kd$ for some integer $k \geq 3$, by Lemma 5.4, we have $d_2(A) < d_2(B)$, $d_2(C) < d_2(D) \leq d_2(E)$, $d_2(F) < d_2(G) \leq d_2(H)$, and $d_2(I) < d_2(J) \leq d_2(K)$. Also,

$$\begin{aligned} d_2(K') &= d - \lfloor d/2 \rfloor + 1 = \lceil d/2 \rceil + 1 \leq 2 + \lfloor d/2 \rfloor \\ &\leq N/d - 1 + \lfloor d/2 \rfloor = d_2(J'). \end{aligned}$$

Thus, we compare the distances $d_2(B)$, $d_2(E)$, $d_2(H)$, $d_2(K)$, and $d_2(J')$ to find the maximum. Since $d \geq 3$, we obtain $d_2(E) < d_2(H) < d_2(J')$ and $d_2(B) < d_2(K) \leq d_2(J')$. Therefore, $height(T_2^+) = d_2(J') = N/d - 1 + \lfloor d/2 \rfloor$. $\square$

6 CONCLUDING REMARKS

The use of multiple independent spanning trees for data broadcasting in networks provides a number of advantages, including increased fault-tolerance and bandwidth, when the connectivity of the underlying network permits such trees to be defined. Moreover, the presence of independent spanning trees with lower height will be helpful for reducing the number of steps in data transmitting. Hence, in the design of routing schemes for data networks, constructing multiple independent spanning trees with reduced heights is an important issue. Compared with Iwasaki et al.'s algorithm for constructing four independent spanning trees in chordal rings, we present a new linear time algorithm that improves the height of spanning trees. Appendix C gives a list of comparisons with different values of N and d .

APPENDIX A

The description of the unique path from root 0 to a node x in spanning tree T_i^* for $i = 1, 2, 3, 4$ under the conditions $[N]_d \neq 0$ and $y > 0$. See Table 1, Table 2, Table 3, and Table 4.

APPENDIX B

The description of the unique path from root 0 to a node x in spanning tree T_i^+ for $i = 1, 2, 3, 4$ under the condition $[N]_d = 0$. See Table 5, Table 6, Table 7, and Table 8.

APPENDIX C

The comparison in the heights of spanning trees for chordal rings $CR(N, d)$ with $31 \leq N \leq 48$ and $2 < d < N/3$, where each cell contains the values $height(T_i^*)/height(T_i)$ if $[N]_d \neq 0$ and $height(T_i^+)/height(T_i)$ otherwise for $i = 1, 2$. See Table 9 and Table 10.

REFERENCES

- [1] B.W. Arden and H. Lee, "Analysis of Chordal Ring Network," *IEEE Trans. Computers*, vol. 30, pp. 291-295, 1981.
- [2] J.C. Bermond, F. Comellas, and D.F. Hsu, "Distributed Loop Computer Networks: A Survey," *J. Parallel and Distributed Computing*, vol. 24, pp. 2-10, 1995.
- [3] N. Chalamaiiah and B. Ramamurthy, "Finding Shortest Paths in Distributed Loop Networks," *Information Processing Letters*, vol. 67, pp. 157-161, 1998.
- [4] J. Cheriyan and S.N. Maheshwari, "Finding Nonseparating Induced Cycles and Independent Spanning Trees in 3-Connected Graphs," *J. Algorithms*, vol. 9, pp. 507-537, 1988.
- [5] S. Curran, O. Lee, and X. Yu, "Finding Four Independent Trees," *SIAM J. Computing*, vol. 35, pp. 1023-1058, 2006.
- [6] D.Z. Du, D.F. Hsu, Q. Li, and J. Xu, "A Combinational Problem Related to Distributed Loop Networks," *Networks*, vol. 20, pp. 173-180, 1990.
- [7] P. Erdős and F.D. Hsu, "Distributed Loop Network with Minimum Transmission Delay," *Theoretical Computer Science*, vol. 100, pp. 223-241, 1992.
- [8] A. Itai and M. Rodeh, "The Multi-Tree Approach to Reliability in Distributed Networks," *Information and Computation*, vol. 79, pp. 43-59, 1988.
- [9] Y. Iwasaki, Y. Kajiwar, K. Obokata, and Y. Igarashi, "Independent Spanning Trees of Chordal Rings," *Information Processing Letters*, vol. 69, pp. 155-160, 1999.
- [10] K. Mukhopadhyaya and B.P. Sinha, "Fault-Tolerant Routing in Distributed Loop Networks," *IEEE Trans. Computers*, vol. 44, pp. 1452-1456, 1995.
- [11] L. Narayanan and J. Opatrny, "Compact Routing on Chordal Rings of Degree 4," *Algorithmica*, vol. 23, pp. 72-96, 1999.
- [12] B. Parhami and D.M. Kwai, "Periodically Regular Chordal Rings," *IEEE Trans. Parallel and Distributed Systems*, vol. 10, pp. 658-672, 1999.
- [13] C.K. Wong and D. Coppersmith, "A Combinatorial Problem Related to Multimode Memory Organizations," *J. ACM*, vol. 21, pp. 392-402, 1974.
- [14] A. Zehavi and A. Itai, "Three Tree-Paths," *J. Graph Theory*, vol. 13, pp. 175-188, 1989.
- [15] G.W. Zimmerman and A.H. Esfahanian, "Chordal Rings as Fault-Tolerant Loops," *Discrete Applied Math.*, vol. 37/38, pp. 563-573, 1992.



Jinn-Shyong Yang received the BS degree in electrical engineering in 1993 and the MS degree in information management in 2000, both from the National Taiwan University of Science and Technology. He is presently a doctoral candidate in information management at National Taiwan University of Science and Technology and a lecturer in the Department of Information Management at the National Taipei College of Business. His current research interests include graph theory and parallel computing.



Jou-Ming Chang received the BS degree in applied mathematics from the Chinese Culture University (June 1987), the MS degree in information management from the National Chiao Tung University (June 1992), and the PhD degree in computer science and information engineering from National Central University (June 2001). He is currently a professor in the Department of Information Management at the National Taipei College of Business (NTCB). His

research interests include algorithm design and analysis, graph theory and combinatorics, and parallel and distributed computing.



Yue-Li Wang received the BS and MS degrees from the Information Engineering Department at Tam-Kang University in 1975 and 1979, respectively. In 1988, he received the PhD degree in information engineering from the National Tsing-Hua University. Now, he is a professor and the head of the Department of Computer Science and Information Engineering at the National Chi-Nan University. His research

interests include graph theory, algorithm analysis, and parallel computing.



Shyue-Ming Tang received the MS degree in 1986 from the National Defense Management College. In 2002, he received the PhD degree in information management from the National Taiwan University of Science and Technology. He is presently an associate professor at the National Defense University. His current research interests include graph theory and algorithm analysis.

► **For more information on this or any other computing topic, please visit our Digital Library at www.computer.org/publications/dlib.**